

Modeling Stylus Dynamics with High-Speed Photography

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Bachelor's Thesis

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A handwritten signature in black ink, appearing to read "Marjorie Shapiro".

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Abstract

The research for this thesis was conducted from fall 2022 to spring 2024 at Project IRENE's office in Lawrence Berkeley National Laboratory. This project was conceived as an effort to understand how styli move on cylinder record surfaces in order to improve audio created from optical reconstruction by IRENE. This thesis details the various approaches that were designed to achieve this end. Background information is presented to contextualize the various challenges there are to understanding stylus motion on cylinder phonographs. Then details about the approach of high-speed photography audio, measurement of cylinder cutter angle, and piezoelectric actuator systems are discussed.

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1. Introduction

The history of audio recording details the evolution of physical processes that transcribe sound. The first known recording of a human voice was captured in 1860 by Édouard-Léon Scott de Martinville. He achieved this through his invention called the ‘phonautograph’, which inscribed an image of his voice onto paper using a horn attached to a membrane that vibrates a stylus. That stylus inscribed a line onto soot-blackened paper. Though the phonautograph lacked playback capabilities, the principle mechanisms of audio recording were already there. By 1877, Thomas Edison had revised and improved this horn-membrane-stylus system and invented the phonograph which could record and playback tinfoil cylinders. In the decades that followed, cylindrical recordings made of wax were the most commercially available.

Grooved audio recordings of this type are susceptible to damages such as mold, dust, cracks, and scratches. The playback stylus also wears down the grooves from use, especially for earlier recordings made from softer materials. For this reason, LBNL developed IRENE (which stands for Image, Reconstruct, Erase Noise, Etc.), a system which can digitize recordings into 2D or 3D images using high-resolution scanners. These images can be analyzed and processed into audio. In this way, audio can be reconstructed from recordings that are too delicate or broken to be played by a playback stylus again.

This research focuses on stylus motion on cylindrical records. The common way of retrieving information about how a stylus moves horizontally and vertically on a monophonic record is through a stereophonic magnetic cartridge. The mismatch of a stereophonic stylus cartridge being used on a mono recording can produce a distorted vision of how the stylus is behaving. A high-speed camera can observe exactly how a stylus moves and theoretically can model the horizontal and vertical motion of a stylus in an independent way other than using a stereophonic stylus cartridge. The way IRENE processes images of grooves into audio relies on an understanding of how styli respond to those grooves, so a system that tracks stylus motion can improve IRENE audio fidelity and help develop physical models of stylus response. A method of optically reproducing audio from cylinder phonographs with high correlation to stereophonic stylus audio is developed using a Chronos 1.4 high-speed camera. The amount of

information in each component of stylus movement on cylinders with complex auditory signals like music or speech are calculated by two methods: signal to noise ratio and correlation. A minutes-long reconstruction of an Edison Blue Amberol Recording of “Sailor’s Hornpipe Medley” reveals that a sizable amount of information is contained within the horizontal motion of stylus motion, in contrast to the notion that information is stored solely in the vertical modulation of cylindrical phonograph grooves. This suggests that the cutter angle, the angle at which a cutter engraves the groove into phonograph cylinders, deviates from the normal of the record surface. Measurements of the cutter angle for one cylinder from high-speed photography audio are $11.8^\circ \pm 0.5^\circ$ while the angle from IRENE audio is $9.4^\circ \pm 0.2^\circ$. Finally, an approach to modeling stylus responses to different stimuli using a piezoelectric actuator is designed.

2. Background

2.1 Cylinder Phonographs

Cylinder phonographs were the first commercially available form of audio recording and remained in production up until the 1920s, when their popularity fell to disc records. Cylinders were made from a variety of materials, most commonly wax or cellulose [1]. They were recorded by a cutter connected to a vibrating diaphragm. The cutter made a vertical groove into the medium, and by replacing the cutter with a stylus the recording could be played back. Cylinders were most commonly cut at 100 TPI (tracks per inch) or 200 TPI and could hold 2 to 4 minutes of audio. Figure 1 shows an Edison Blue Amberol Cylinder and cylinder container.



Figure 1: The Right of the People to Rule / Teddy Roosevelt. Edison Blue Amberol: 3707. 1919, Courtesy of UCSB Cylinder Audio Archive [2]

2.2 Groove Geometry

When grooves are cut into a cylinder, it spins about its axis of symmetry and translates laterally. In this manner, the recording takes the form of an uninterrupted helix across the cylinder's surface. Cylinder recordings are monophonic, meaning the cutter engraves one channel of audio into the medium. Additionally, the grooves are cut medially from the surface towards the axis of rotation. In ideal conditions, the information transmitted to the cutting stylus should be entirely contained in the vertical component of the groove [3]. Likewise, all lateral modulations in the groove should entirely be noise. In actuality, the cutter engraves the groove at an angle from the normal of the cylinder's surface, resulting in information being stored both in the vertical and horizontal groove modulations. Figure 2 shows the cross section of vertical grooves on a cylinder's surface.

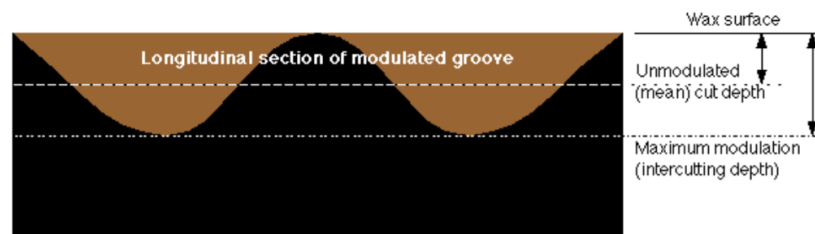


Figure 2: Cross sectional diagram of vertical grooves on a cylinder's surface, courtesy of POPPY Records [4]

The frequency response of the cylinder is limited by the geometry of the cutting stylus and playback stylus. For most cylinders, the signal response is weak for frequencies above 4 to 5 kHz. For comparison, the human hearing range is about 20 Hz to 20 kHz. Playback styli are rounder than cutting styli. Consequently, playback styli follow a different path along grooves than the paths carved by cutters, a phenomenon known as tracing distortion. If the playback stylus does not match the angle at which the cylinder was cut, the tracing distortions can become more pronounced [5]. Figure 3 shows the different paths that a chiseling stylus and playback stylus would take on a cylinder surface as a result of the different shapes of their tips.

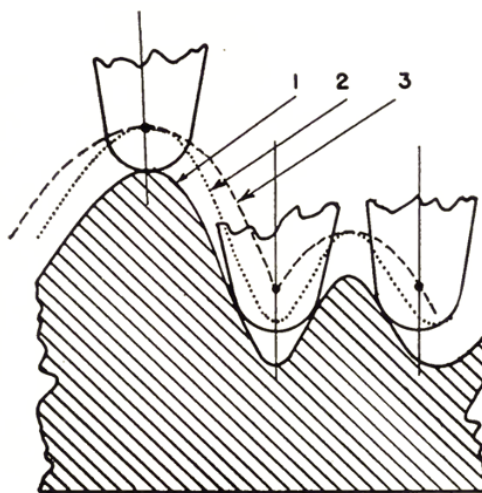


Figure 3: Tracing Distortion. (1) Recorded undulation, (2) parallel curve to curve 1, (3) path of reproducer stylus, Courtesy of M. J. Di Toro [6]

2.3 The 45/45 System

Cylinder phonographs predate the invention of stereo recording, or the method of recording two channels of audio into a single groove. Stereo recording for disc records commonly uses the 45/45 system, with each wall of the groove cut 45° from the disc's surface normal [4]. One wall of the disc's groove contains left channel audio, while the other wall contains right channel audio. The vertical and horizontal components of audio with respect to the 45/45 system are depicted in figure 4.

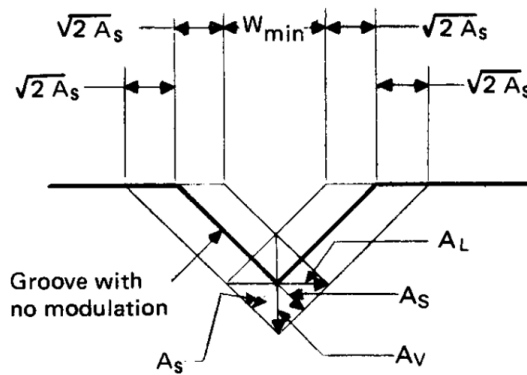


Figure 4: Cross section of a groove from a 45/45 stereo record, courtesy of Erling P. Skov. [7]

2.4 Stereophonic Stylus Cartridges

Stereophonic styli cartridges play stereo discs with the 45/45 system. When listening back to a recording, the groove walls move the stylus tip. This mechanical movement is translated through a lever arm to a magnet, which then transduces two electrical signals to each of two coils oriented 45° degrees from the magnets normal in a V-shape. Thus, the mechanical movements produced by each groove wall on the stylus are electromagnetically separated [4].

When studying the vertical and horizontal components of audio from a cylinder phonograph, the most readily available option is to use a stereophonic stylus cartridge designed for the 45/45 system of disc records. Indeed, a stereo cartridge, either 45/45 or 0/90 is a more general way to study a mono recording and grants the freedom to combine the 2 signals in any way we can choose.

$$A_V = A_L - A_R$$

$$A_H = A_L + A_R$$

Equation 1: Combination of left and right audio in order to create vertical and horizontal audio.

A_V is vertical audio, A_H is horizontal audio, A_L is left audio, A_R is right audio

2.5 IRENE Scanning

IRENE digitizes historical audio artifacts through high-resolution 3D imaging of record surfaces. Then through a chain of image and signal processing, the audio from the record is optically reproduced. The scanning probe and software used to reproduce audio from cylinders used in this project are detailed in this section.

2.5.1 CLS Probe

Cylinders in this project are scanned by a CHRcodile Chromatic Line Scanner (CLS) manufactured by Precitec. The CLS probe uses color coded confocal microscopy to scan the depth of a line across a cylinder' surface [8]. The cylinder slowly rotates for a full revolution as the CLS slowly scans before translating by a small amount. This repeats until the entirety of the cylinder is scanned, a process which requires a few hours. Figure 5 is a picture of the CLS probe used to scan 'Sailor's Hornpipe Medley.'



Figure 5: CLS probe on IRENE optical bench

2.5.2 Weaver

Weaver is a software program developed by IRENE to detect groove paths as imaged by the CLS probe. It searches for the path a stylus would take by drawing lines through the deepest parts of the groove. Because the cylinder surface appears as a rectangle on Weaver, software attempts to connect groove paths that go off the end of one side of the screen and reappear on the other side. This reproduces the long uninterrupted groove on the surface of a cylinder

phonograph. Because the CLS probe samples the cylinder's surface at a higher rate than a playback stylus, unfiltered audio from Weaver carries a lot of high frequency noise that would not be produced by a physical stylus. To increase audio fidelity, Weaver has noise reduction features to rid of this high frequency signature and other forms of noise. Additionally, Weaver has some restorative functions that reduce noise from certain damages on record surfaces such as dust, scratches, and mold. Figure 6 shows the CLS scan of 'Sailor's Hornpipe Medley' loaded on the front panel of Weaver.

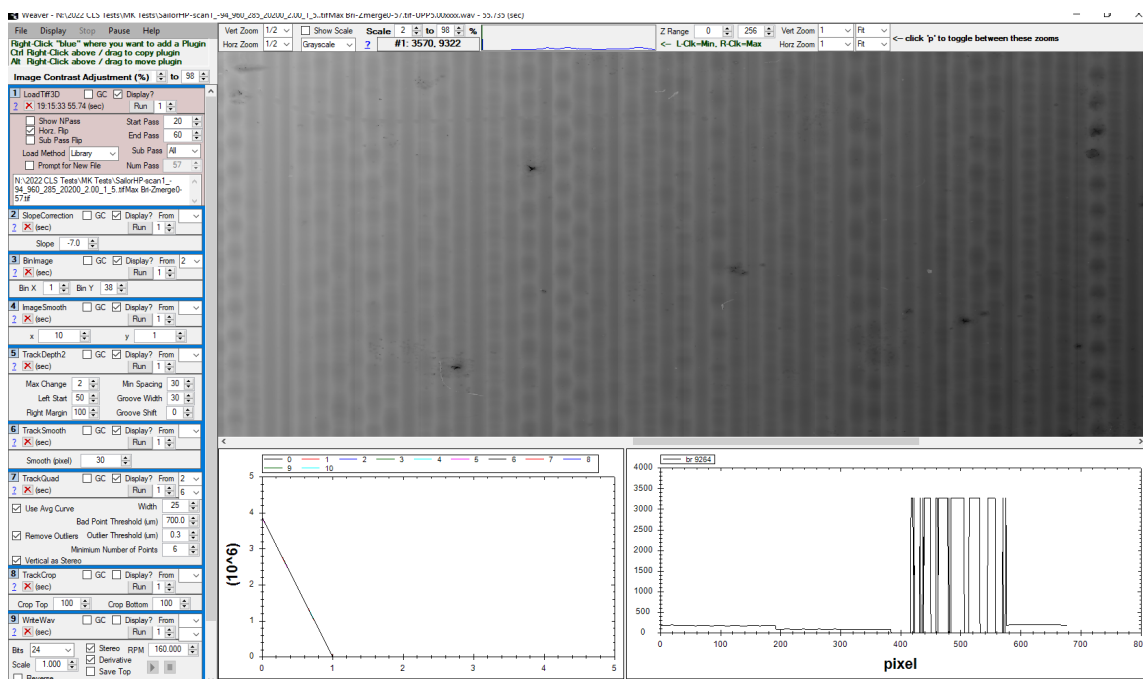


Figure 6: A CLS scan of 'Sailor's Hornpipe Medley' inside Weaver

3. High-Speed Photography Audio

3.1 Chronos 1.4

The Chronos 1.4 is a high-speed camera used in this experiment to capture audio information from the movement of styli and piezoelectric actuators. It is capable of photographing at 1,069 fps at its widest-screen setting of 1280x1024 pixels, and has a maximum

frame rate of 40,413 fps at its narrowest resolution. Figure 7 shows the Chronos 1.4 on the lab bench of IRENE.

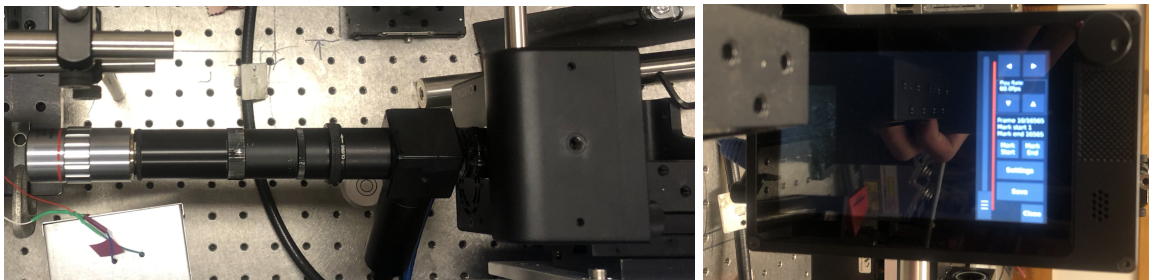


Figure 7: Pictures of Chronos 1.4 (Left) Top down view of Chronos 1.4 camera with tube extensions, (Right) Front-facing screen of Chronos 1.4

The frequency response of audio extracted from stylus and piezoelectric movement has an upper bound set by the Nyquist frequency of the photography frame rate. Since the frequency response on cylinder phonographs is 4-5 kHz, a frame rate between 8-10kHz suffices to capture the signal from stylus movement.

$$\text{Frequency Response} = \frac{1}{2} \text{Frame rate}$$

Equation 2: Nyquist frequency of high speed photography audio

3.2 Focusrite 2i2

The Focusrite 2i2 is an audio interfacing device that transmits up to 2 analog audio signals to a computer through a USB port. The stereophonic stylus cartridge is connected to the Focusrite 2i2 with the left channel audio connected to one input and the right channel audio connected to the other. The gain for each input is adjusted by a single-turn potentiometer on the Focusrite 2i2's front panel. Using a function generator, the gains are matched to $\pm 0.1\text{dB}$ with a signal on the same order of amplitude as stylus audio. When in recording mode, the computer stores the incoming signals in a stereo audio file. Figure 8 shows the Focusrite 2i2 used to record audio from the stereophonic cartridge.



Figure 8: Focusrite 2i2 audio interface

3.3 Stylus and Cylinder Control

The stereophonic stylus cartridge is connected to a lever arm that is raised and lowered by a remote-controlled motor. The motor can change the stylus arm position via commands in an IRENE software program named 'I.' The I program also controls the rotation and translation of a mandrel connected to a rotary stage. In this manner, a cylinder attached to the mandrel can be spun and translated at the right rates for proper playback with the stylus dropped onto its surface. The spin and translation rate used to move the cylinders used in this experiment are tabulated below.

Cylinder Track Density (TPI)	Spin Rate ($^{\circ}/s$)	Translation Rate (m/rot)
100	960	0.00070555
200	960	0.00035277

3.4 Chronos 1.4 Control

Recording, lighting, and cylinder motion start-up are controlled through a Labview virtual instrument named 'Chronos_Linear.vi.' The software is connected to the Chronos 1.4, a

light source, the motor that controls the stylus lever arm, and the mandrel and rotary stage that move the cylinder. When 'Chronos_Linear.vi' is run, it performs a series of actions that allow for a quick start-up for recording. It drops the playback stylus onto the cylinder, accelerates the cylinder, waits for the cylinder to reach the right velocity, turns on the light source to the desired brightness, and then enables recording mode on the Chronos 1.4 for the desired duration.

3.5 Experimental Design

In order to study stylus movement on cylinder surfaces, the stylus needs to be simultaneously filmed by the Chronos 1.4 and recorded by the Focusrite 2i2. The Focusrite 2i2 is recorded enabled and then Chronos_Linear.vi is run so the stylus plays the cylinder and the Chronos 1.4 begins recording. The photographs from the Chronos 1.4 are then saved to a network drive and the Focusrite 2i2 audio is saved to local computer storage. A schematic of the experimental design and an in-lab picture of the setup are shown below in figure 9.

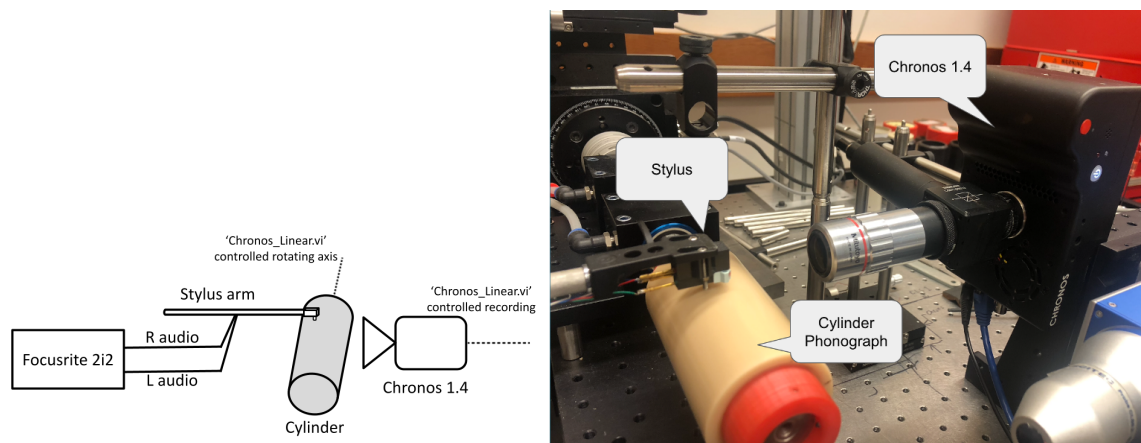


Figure 9: Experimental design (Left) Schematic of experimental design, (Right) In-lab picture of experimental setup

In this research, the Chronos 1.4 is set at 8492 fps with a resolution of 640x300 pixels. After rotating the camera 90° onto its side, these parameters allow for the stylus to always remain in picture while moving on the cylinder. Additionally, the frame rate is high enough to capture the cylinder's audio information from the stylus's movement. Figure 10 shows a typical picture of the stylus tip and cantilever as images by the Chronos 1.4.

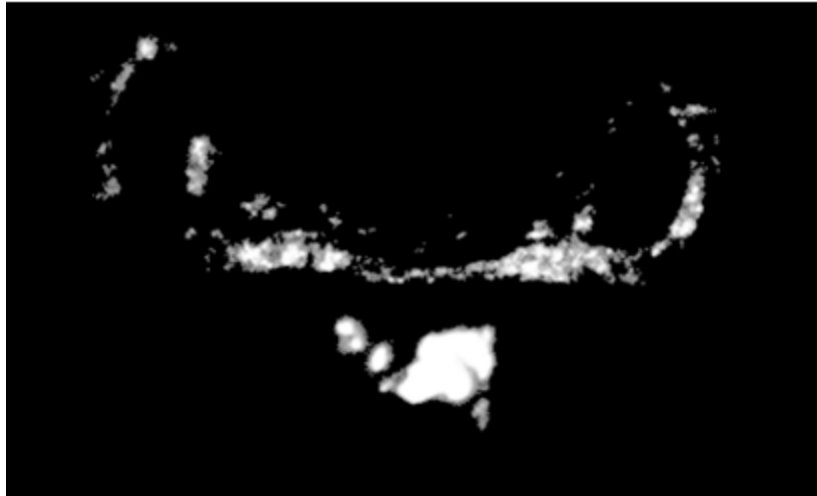


Figure 10: Image of a stylus tip and cantilever from the Chronos 1.4

3.6 Image to Audio Processing

3.6.1 Pattern Tracking

The frame-by-frame position of the stylus was tracked using image processing tools inside National Instruments Vision Assistant. The raw photographs from the Chronos 1.4 are passed through a smoothing filter to help rid of noise. Then they are rotated 90° to restore the upright orientation of the stylus onto the cylinder surface. At this point, a single pattern template is made from the first photograph of each recording. The pattern template stores the strongest curves and edges from the photograph. Vision Assistant searches for the best matching pattern to the template in each image and records the position where it found the match. This method reliably finds a match in >99% of images in each recording. Figure 11 shows the front panel of the Vision Assistant image processing chain used for pattern matching.

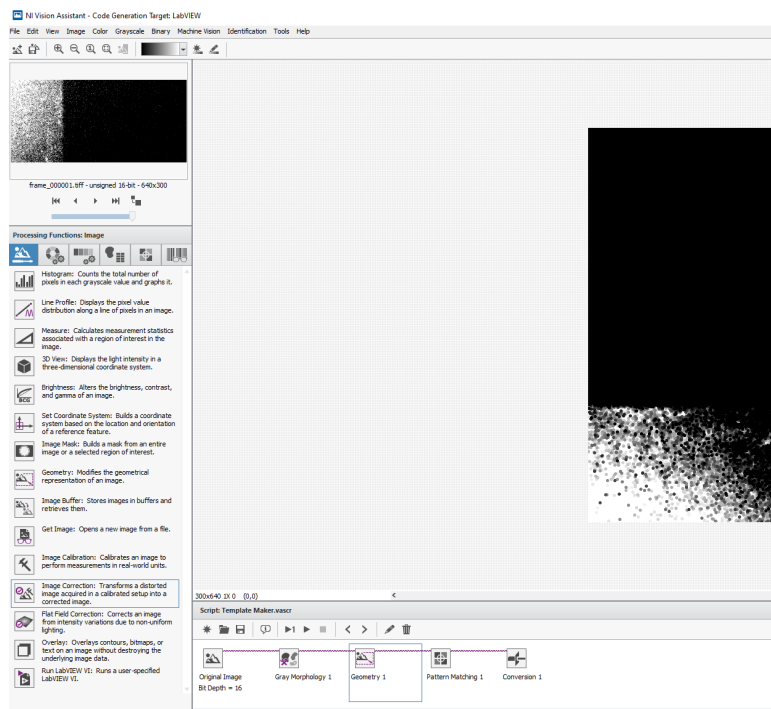


Figure 11: Vision Assistant image processing chain

An alternative method of using the centroid of each image to track stylus position was attempted. The centroid method has the advantage that it works for 100% of the images in each recording. However, the Chronos 1.4 captures light that reflects off the cylinder in addition to light from the stylus. The light from the cylinder affects the position of the centroid and thus the results are inconsistent with the actual stylus movement.

3.6.2 Image_to_wav.vi

The pattern tracking routine from Vision Assistant is implemented inside a Labview virtual instrument named 'image_to_wav.vi.' This routine charts both the vertical and horizontal position of the pattern match from each image in a recording. Since the sound picked up by the stylus cartridge is proportional to the stylus' velocity, audio can be recreated from the video by taking the first order discrete derivative of the stylus position. For the <1% of images that did not have a template match, the velocity is interpolated. Figure 12 shows the front panel of image_to_wav.vi.

$$\frac{f_{i+1} - f_i}{h} \approx f'(x_i)$$

Equation 3: First order discrete derivative

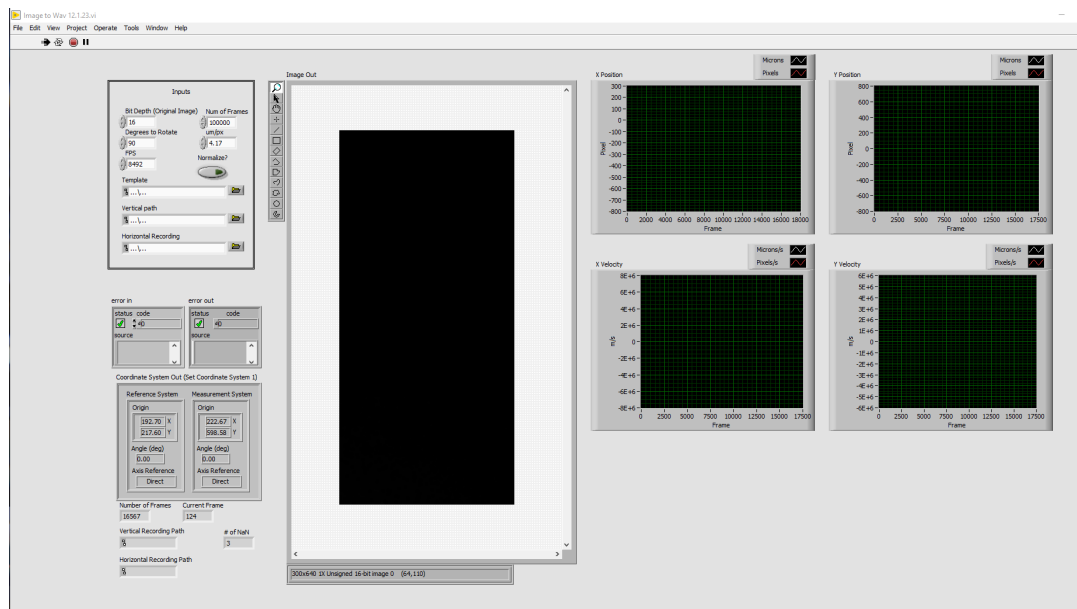


Figure 12: Pattern tracking using image_to_wav.vi

3.6.3 Reproduced Audio vs. Analog Audio

In order to test the fidelity of audio as produced by the pattern matching method, the audio is synchronized with analog audio from the Focusrite 2i2 and a correlation coefficient is calculated by the Pearson method. An initial correlation is calculated after matching up the waveforms by eye, however this is not always sufficient. Additional correlation coefficients are calculated by lagging one audio file one sample at a time in order to find the highest correlation, as shown in figure 13.. The results demonstrate that the audio produced from the Chronos 1.4 is highly correlated to the audio from the Focusrite 2i2, suggesting the pattern tracking method reproduces audio from stylus movement with high accuracy. Figure 14 shows the correlation between the Chronos audio and the Focusrite 2i2 audio at the best lag over a 10 second snippet of audio.

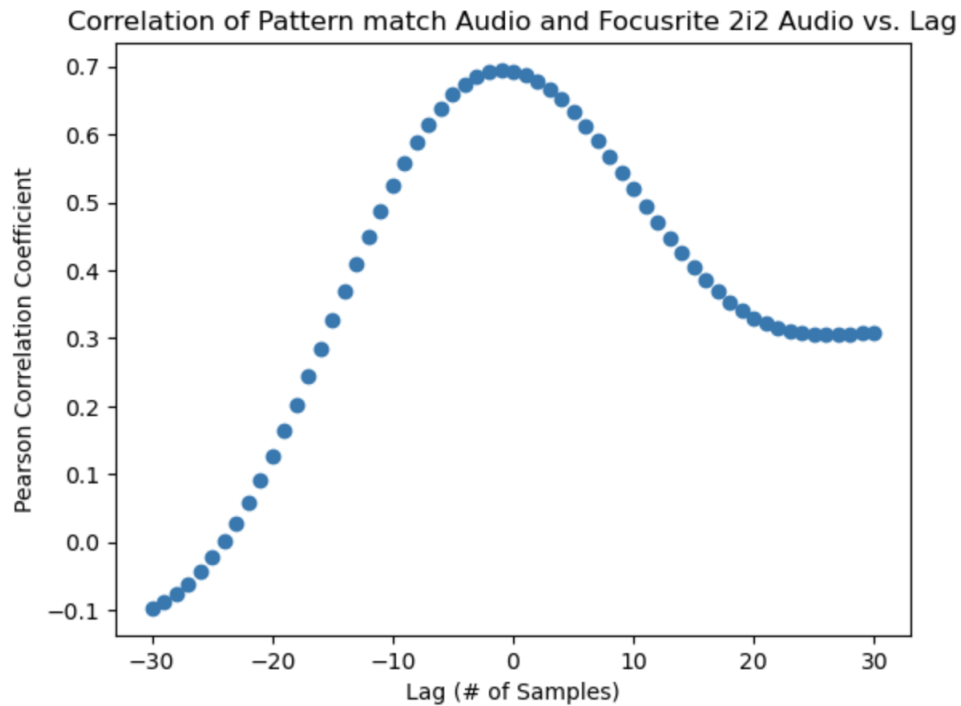


Figure 13: Correlation between audio created from Pattern Tracking and Focusrite 2i2 Audio vs. lag. Audio is from a snippet of a cylinder titled “Stars and Stripes Forever.” The maximum correlation is 0.69 at a lag of -1 samples.

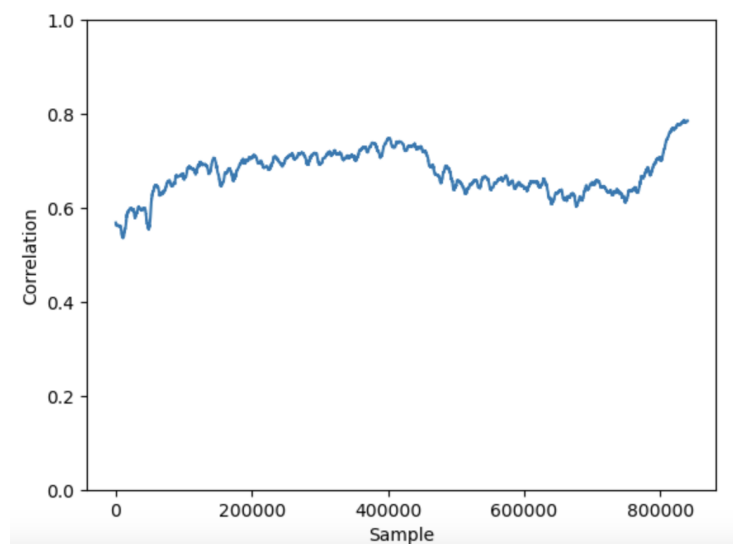


Figure 14: Correlation vs. Time for the snippet of audio from “Stars and Stripes Forever.” One of the audio files is lagged by -1 samples.

4. Measuring Cutter Angle

The cutter angle describes the lateral angle from the surface normal at which a cutting stylus engraves a groove into a cylinder, as shown in figure 15. If a cutting stylus engraves grooves into a cylinder towards its axis of rotation, all of the information should be carried in the vertical motion of the playback stylus. Consequently, any horizontal stylus motion should be noise. If the cutting stylus cuts the cylinder at an angle, then the playback stylus movement will carry the maximum amount of information at that same angle. The component of stylus movement orthogonal to this direction of maximum information should be entirely noise.

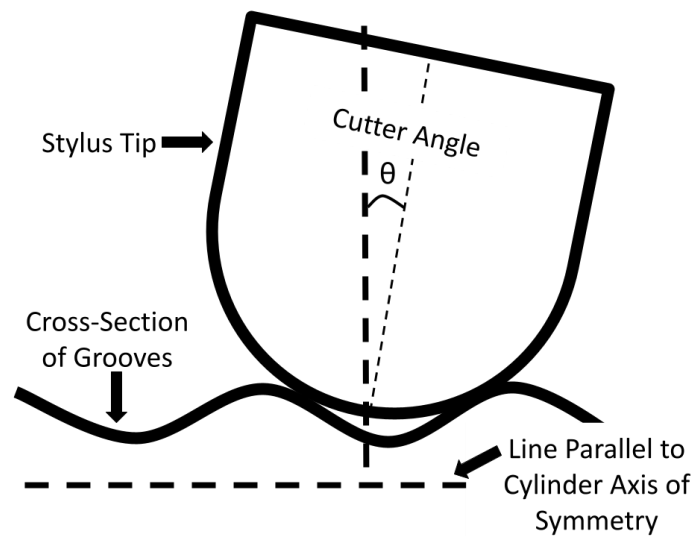


Figure 15: Diagram of Stylus Cutter Angle

4.1 Rotating Audio

The pattern matching method produces two audio files, one from the vertical movement of the stylus and the other from the horizontal movement. Likewise, the Focusrite 2i2 produces left and right audio files that can be recombined to make vertical and horizontal files. Additionally, Weaver can use images from the CLS probe to construct audio files from both the vertical and horizontal modulations of the grooves. These vertical and horizontal audio files can be transformed into any two orthogonal components of the stylus' motion through the rotation operator.

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} H \\ V \end{pmatrix} = \begin{pmatrix} H' \\ V' \end{pmatrix}$$

Equation 4: Rotating Audio. H and V are horizontal and vertical audio files, H' and V' are rotated horizontal and vertical audio files

4.2 Test Cylinder: ‘Sailor’s Hornpipe Medley’

An Edison Blue Amberol cylinder titled ‘Sailor’s Hornpipe Medley’ is used as the test cylinder for this portion of the experiment. It is a 100 TPI recording approximately 2 minutes long. It features a vocal introduction followed by an instrumental song arrangement that lasts all the way until the cylinder’s conclusion. The cylinder has a scratch on it that produces a repetitive clicking noise every time the stylus passes over it for between the 0:15 and 0:48 marks of the pattern matching and Focusrite 2i2 audio. The Chronos 1.4 has a 32GB memory RAM which limits the length of time it can record before pausing to upload all images to the network drive. At 8492 fps, the Chronos 1.4 can only record for 10 seconds at a time. A complete recording of ‘Sailor’s Hornpipe Medley’ made from the pattern matching requires 14 different 10-second recordings.

A more ideal test cylinder for finding the angle of stylus motion where the signal is maximized would be a test tone cylinder with a known signal. This would allow for easy quantification of signal and noise. However, at the time of data collection, there was no test tone cylinder available.

4.3 SNR vs. Correlation Calculations

This research explores two methods for calculating the cutter angle for ‘Sailor’s Hornpipe Medley’ and other cylinders with complex auditory information. In the first method, audio files from the vertical and horizontal directions are segmented into an arbitrary number of audio files of equal length. Then, the vertical and horizontal components of each segment are rotated a number of times across a list of discrete angles. At each angle, the RMS amplitude is calculated for both the rotated vertical and horizontal files. The signal to noise ratio (SNR) is

taken to be the ratio of the RMS rotated vertical amplitude to the RMS rotated horizontal amplitude. The SNR is plotted for each segment of audio across the range of angles, as shown in figure 16. The cutter angle is calculated by averaging the angles of maximum SNR across all segments of audio.

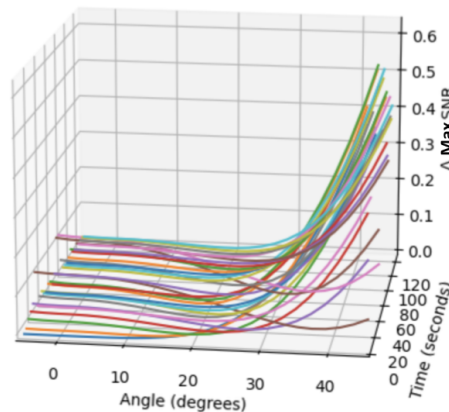


Figure 16: Difference from Maximum SNR vs. Time vs. Angle for Focusrite 2i2 Audio

The second method of calculating the cutter angle finds the angle at which the correlation between rotated vertical and horizontal audio files is minimized. Similar to the SNR method, the correlation method segments the audio and rotates each segment through a series of angles. At each angle, a correlation coefficient is calculated between the rotated vertical and horizontal audio using the Pearson correlation method. The correlation coefficient is plotted for each segment of audio across all angles, as shown in figure 17. Then, the cutter angle is calculated by averaging the angles of minimum correlation across all segments of audio.

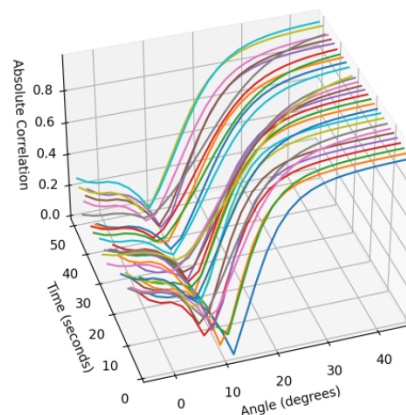


Figure 17: Absolute Correlation vs. Time vs. Angle for Focusrite 2i2 Audio

4.4 Results

The standard deviation of cutter angles for each segment of audio is consistently higher for the SNR method than the correlation method. This could be due to the fact that the SNR method assumes that the rotated vertical audio is pure ‘signal’ and the rotated horizontal audio is pure ‘noise’ at the cutter angle. However, some amount of music from “Sailor’s Hornpipe Medley” is discernible no matter the angle in the rotated horizontal audio. This suggests that cylinders are not cut so that all information is contained in a single direction. The correlation method finds the angle at which the rotated vertical and rotated horizontal audio files are most dissimilar, which should still be the case at the cutter angle.

The results from the correlation method on a 1 minute segment of “Sailor’s Hornpipe Medley” are shown in the table below and in figure 18. The results from the Focusrite 2i2 are far apart from the Chronos 1.4 and CLS probe, likely because of the use of a stereophonic cartridge to play a mono recording. The cutter angle between the Chronos 1.4 (11.81°) and CLS probe (9.37°) are close but not within error of one another. The results from the Chronos 1.4 can be skewed if the Camera is misaligned with the playback stylus.

Method	Cutter Angle
Chronos 1.4	$11.81^\circ \pm 0.48^\circ$
Focusrite 2i2	$17.91^\circ \pm 0.29^\circ$
CLS Probe	$9.37^\circ \pm 0.18^\circ$

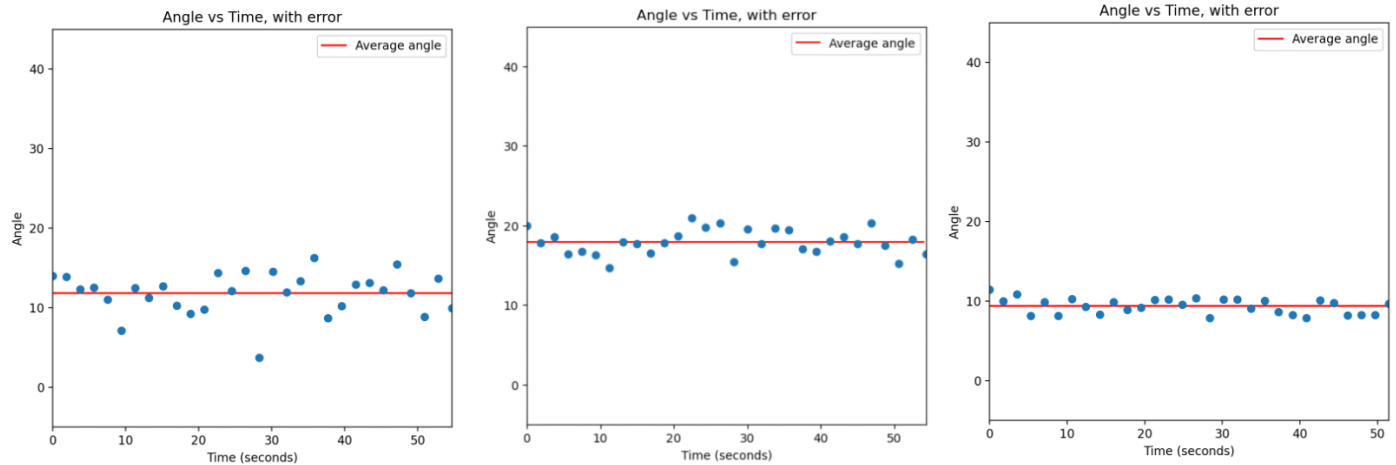


Figure 18: Cutter Angle vs Time plots. (Left) Results from Chronos 1.4 Audio, (Center) Results from Focusrite 2i2 Audio, (Right) Results from CLS Audio. Error bars are small, obscured by data points.

5. Piezoelectric Actuator

5.1 System Design

Piezoelectric components move in a well-defined manner, thus they make a good choice to calibrate the Chronos 1.4 tracking system and to study how styli behave in response to different stimuli. The piezoelectric system designed for this research uses an AE0505D44H40DF piezoelectric stack actuator, as pictured in figure 19. It has a capacitance of 3.4 μF and a maximum displacement of 44 μm at 150 V. For comparison, the modulations of grooves on cylinders have an amplitude of approximately 10 μm , meaning that the piezoelectric can simulate the full range of motion a stylus would experience on a cylinder with approximately 50 V.

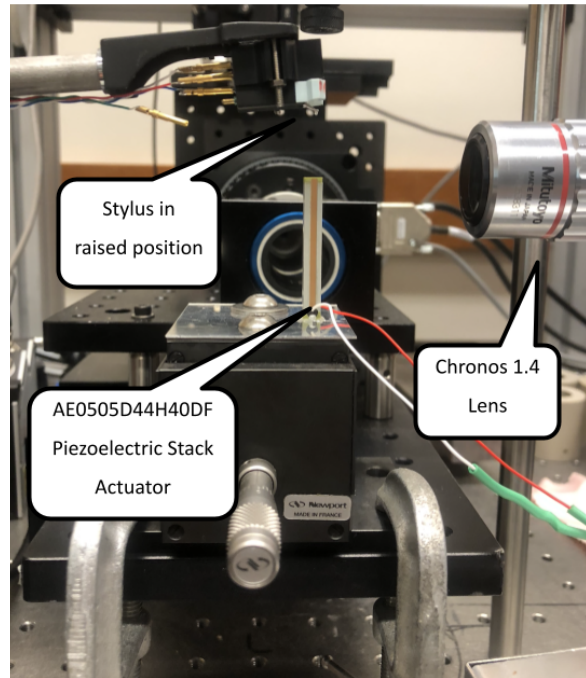


Figure 19: Piezoelectric setup in front of Chronos 1.4

The signal for the piezoelectric is generated by a function generator and then amplified by a PDu150CL piezo driver. The amplifier has a maximum output of 150 V at 300 mA. The output of the amplifier is connected to a 1 k Ω resistor with a 20 W power rating. These resistors are then connected to a switch that either powers the piezoelectric actuator or shorts to ground. While the piezoelectric oscillates, the Chronos 1.4 photographs and records its movement, as shown in figure 20.

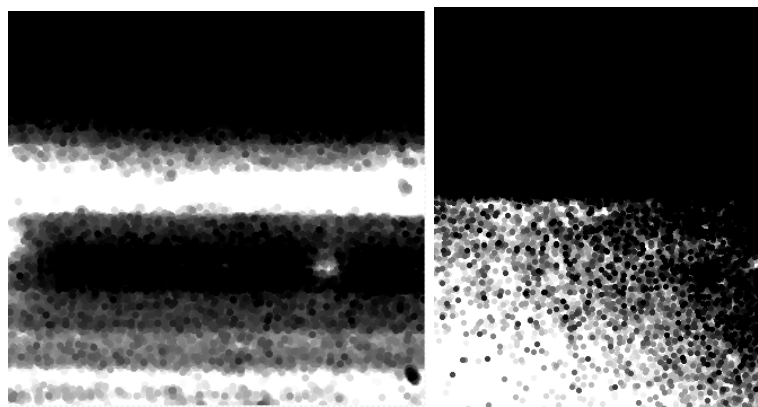


Figure 20: Chronos 1.4 pictures of piezoelectric actuator. (Left) Image of the piezoelectric with low magnification, (Right) Image of the piezoelectric with high magnification

5.2 Preliminary Results

The Chronos 1.4 is able to track piezoelectric movement. Signals up to 100Hz have been recreated by the Chronos 1.4 after low-pass filtering the raw position tracking data from the Chronos, as shown in figure 21.

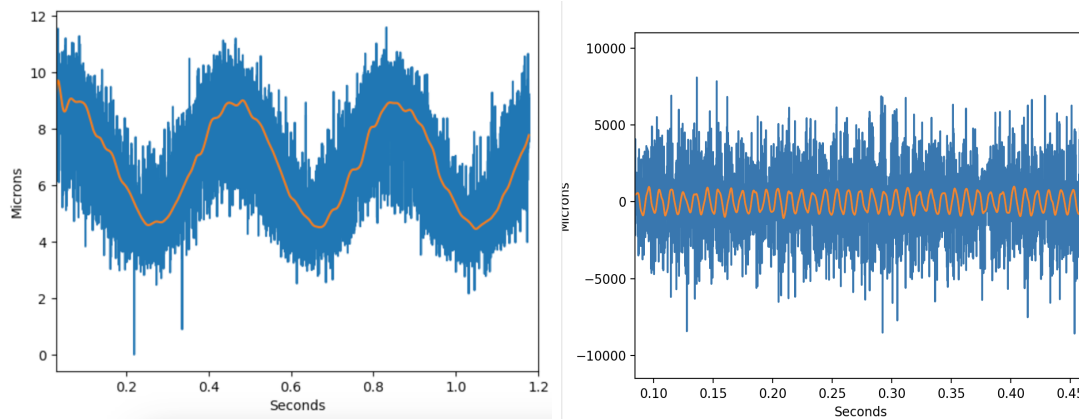


Figure 21: Chronos 1.4 Tracking of piezoelectric movement. (Left) Chronos tracking of the piezoelectric oscillating at 3Hz, (Right) Chronos tracking of the piezoelectric oscillating at 100Hz. In blue is raw data of Chronos 1.4 tracking of stylus position, yellow is the data after low-pass filtering.

The SNR of the Chronos 1.4 tracking the piezoelectric decreases with frequency. In order to improve the SNR, extension tubes were added between the Chronos lens and aperture in order to increase image magnification. This amplifies the magnitude of movement of the piezoelectric on the Chronos 1.4 screen. An effort to recreate piezoelectric signals at this higher magnification was made, however the piezoelectric surface appears more homogenous, increasing the noise in the pattern matching method. Thus, more work is needed to reproduce signals at this higher magnification.

5.3 Future Directions

IRENE has interferometers manufactured by Micro-Epsilon capable of measuring the height of the piezoelectric. Once it is proven that Chronos 1.4 images of the piezoelectric can reproduce signals up to 4-5 kHz, the Chronos 1.4 signal reproduction can be corroborated by interferometer measurements of the piezoelectric displacement. A schematic of this proposed

experimental design is shown in figure 22.. This would confirm the accuracy of the results produced by the Chronos 1.4.

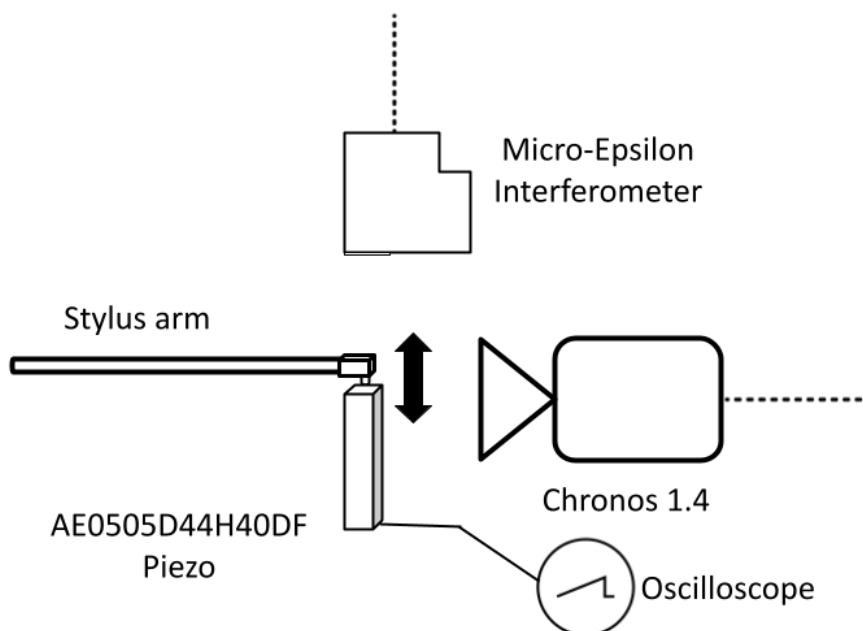


Figure 22: System schematic of measurement of piezoelectric displacement

The end-goal of the piezoelectric system is for it to drive movement of a playback stylus. When the stylus is dropped onto the piezoelectric, the Chronos 1.4 could independently track the displacement of both the piezoelectric and the stylus. In this manner, it could characterize how playback styli respond to signals of any duration and frequency up to 5kHz.

6. Conclusion

The Chronos 1.4 is a versatile and adaptive tool for measuring and modeling how styli move on cylinder phonographs. It also provides an independent method of measuring qualities of stylus movement apart from IRENE's optical scanning. Audio reproduced from high-speed photography of stylus movement has high correlation to the analog audio from stereophonic stylus cartridges, suggesting it is an apt method of recovering information from stylus vibrations. The correlation method of calculating cutter angle proves more robust than the SNR method. The measurement of cutter angle on "Sailor's Hornpipe Medley" suggests that a significant amount of information is contained in the horizontal component of stylus motion,

unlike previously thought. The piezoelectric actuator method of modeling stylus dynamics is in its infancy, but the first results indicate that the Chronos 1.4 should be fully capable of tracking piezoelectric oscillations at frequencies across the audio range of phonographic cylinders. The Chronos 1.4 and piezoelectric together provide exciting new opportunities for research of stylus motion that would be of great benefit to IRENE and the preservation of historic audio artifacts.

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